
Similar Scores, Different High-Risk Episodes: Auditing Model Selection in Longitudinal EHRs

Polina Korobeinikova¹ Dmitry Lvov^{1,2} Ilya Pershin^{1,3}

¹Research Center of the Artificial Intelligence Institute, Innopolis University, Innopolis, Russia

²ITMO University ³Kazan Federal University

Abstract

Clinical prediction pipelines are commonly selected by small differences in aggregate held-out scores, although operational prioritization depends on calibrated risks, retraining stability, and which episodes fill the chosen capacity. Longitudinal EHRs make this problem acute: repeated diagnoses can represent persistent disease or redundant documentation, and compressing them can change both prediction quality and the history visible to a model. We compare Raw histories with a persistence-aware Era+backfill representation on EHRSHOT 30-day readmission and ICU-transfer using a decision-aware evaluation of cohort performance, uncertainty, calibration, retraining, selected sets, and controlled repetition. Across five-seed ensembles at 4,096 events, Era+backfill records the better observed value in all ten conventional task-metric comparisons, although every paired patient-cluster interval includes zero. At the same top-decile capacity, it captures three additional readmission positives and one additional ICU positive while replacing 34 (15.5%) and 51 (25.0%) episodes in each prioritized set. These replacements remain after five-seed ensembling, while within-representation single-seed comparisons also show substantial variability. A structure-matched control locates the clearest ICU gains in explicit persistence state, while controlled diagnosis repetition reverses the sensitivity ordering across tasks. Aggregate rankings therefore neither determine whom a pipeline prioritizes nor separate representation effects from retraining instability. A joint evaluation profile makes those model-selection consequences visible. Code and publication-safe aggregates are available at <https://github.com/LvovDmitro/ehrshot-state-or-space>.

1 Introduction

Clinical risk models influence which records receive attention and which patients are prioritized for follow-up. Yet representation choices are often advanced on a small difference in one held-out metric. This practice assumes that the same choice remains favorable under other clinically relevant metrics, repeated training, calibration assessment, and the threshold used to prioritize cases.

Longitudinal electronic health records (EHRs) make this assumption especially consequential. A patient timeline may contain thousands of irregular events and the same diagnosis repeated across visits. A Raw suffix preserves the recorded stream, whereas a persistence-aware representation can summarize repeated diagnoses as explicit condition states and use the released context positions for earlier history. Comparing these representations with one held-out score captures average prediction quality, but says little about whether they rank the same episodes or respond similarly to repeated documentation.

We examine this gap on the EHRSHOT 30-day readmission and ICU-transfer tasks [Wornow et al., 2023]. Figure 1 previews the central result. Our evaluation profile connects predictive metrics,

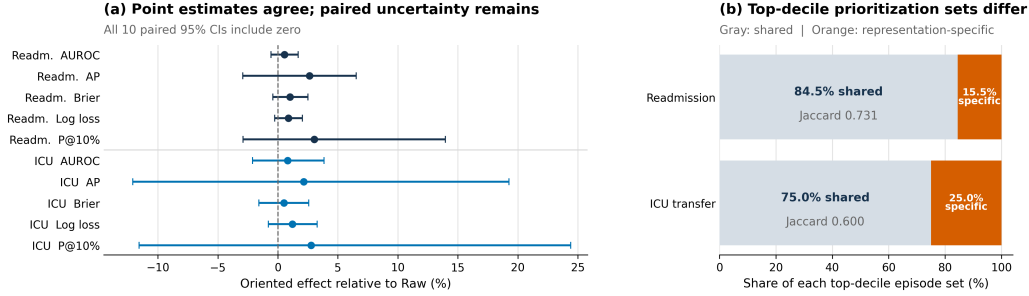


Figure 1: Aggregate direction, paired uncertainty, and top-decile set disagreement. Panel (a) shows oriented effects relative to Raw with paired patient-cluster 95% intervals: all ten point estimates favor Era+backfill, while every interval crosses zero. Panel (b) shows that 15.5% of each readmission top-decile set and 25.0% of each ICU set is representation-specific.

paired patient-cluster uncertainty, calibration, patient weighting, complete retraining, top-decile set agreement, and response to a strictly pre-prediction perturbation. Era+backfill leads all ten primary episode-weighted point estimates, but the paired intervals retain uncertainty and fixed-capacity prioritization differs. Component controls examine explicit state encoding; frozen-pipeline repetition tests probe behavioral sensitivity.

2 Related Work

Longitudinal EHR representations. RETAIN established reverse-time modeling of clinical sequences [Choi et al., 2016]; EHRSHOT later standardized patient-level prediction tasks [Wornow et al., 2023], and MEDS provided a common event-stream format [MEDS Working Group et al., 2024]. *Context Clues* varies context length from 512 to 16,384 tokens on EHRSHOT and studies performance across naturally occurring copy-forward repetition, irregular timing, and disease progression [Wornow et al., 2025]. These studies establish the importance of history length and repeated documentation, but leave two mechanisms entangled under a fixed event budget: encoding repeated diagnoses as persistent state and using positions released by compression to admit earlier history. Consequently, the source of any representation-level difference remains difficult to identify.

Reliable evaluation of clinical prediction. Clinical prediction guidance distinguishes discrimination, calibration, and intended use, and calls for transparent reporting of data splits and model development [Van Calster et al., 2019, Collins et al., 2024]. Average precision (AP) is particularly informative for rare outcomes [Saito and Rehmsmeier, 2015], while problem-aware validation frameworks emphasize metric portfolios rather than a single default score [Maier-Hein et al., 2024]. These practices improve the assessment of a fixed model, but they do not by themselves reveal whether a representation choice is stable across analysis units and retraining.

Underspecification and behavioral stability. Models with similar held-out performance can encode different solutions [D’Amour et al., 2022], and repeated EHR training can alter patient-level risk assignments while preserving aggregate metrics [Lopez-Martinez et al., 2022]. Benchmark reuse can further distort model selection [Dwork et al., 2015, Westphal and Brannath, 2019, Cawley and Talbot, 2010], while copy-forward documentation motivates behavioral tests of repeated information [Thornton et al., 2013]. Together, these strands leave open whether a representation preference remains stable across analysis units, retraining runs, calibration summaries, selected episodes, and structured perturbations.

3 Methodology

We compare Raw and Era+backfill under a common protocol comprising patient-disjoint prediction tasks, fixed task-specific architectures, seed-wise calibration on tuning data, and held-out evaluation at both cohort and top-decile levels. Figure 2 illustrates the representation transformation and shared evaluation pipeline.

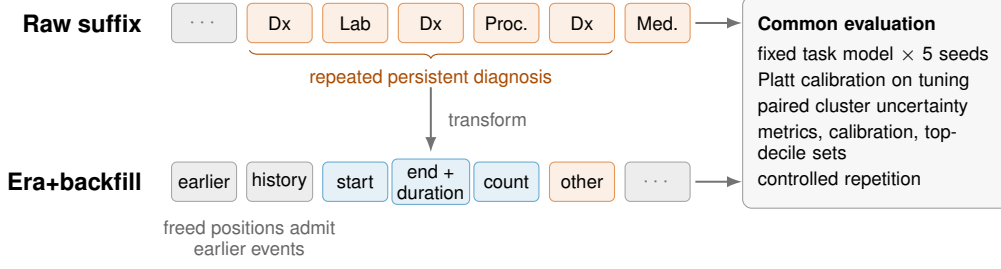


Figure 2: Representation mechanism and common evaluation. Raw preserves repeated diagnosis events. Era encodes start, end-with-duration, and count; compression can admit earlier events under the same L -event budget. Both representations then enter the same task-specific training, calibration, and evaluation protocol.

3.1 Data and prediction tasks

We use EHRSHOT 30-day readmission and ICU-transfer prediction [Wornow et al., 2023, Stanford Shah Lab, 2026]. Readmission is predicted at 23:59 on the discharge day, with outcomes measured over the following 30 days; ICU transfer is predicted at 23:59 on the admission day. The benchmark provides patient-disjoint train, tuning, and held-out assignments. The held-out cohorts contain 2,189 episodes from 1,190 patients for readmission (260 positives) and 2,037 episodes from 1,154 patients for ICU transfer (85 positives); complete split counts appear in Appendix A.

For episode i , let τ_i denote prediction time and g_i its patient. The available history is

$$H_i(\tau_i) = \{(c_{ij}, t_{ij}, x_{ij}) : t_{ij} \leq \tau_i\}, \quad (1)$$

where c_{ij} is an event code and x_{ij} contains optional numeric values. The split audit verified unique episode assignment, patient separation, label coverage, and agreement with the official subject split.

3.2 Persistence-aware histories

We derive the set of candidate persistent diagnoses using only patients assigned to train. A code qualifies if it occurs in at least 50 training patients and at least half have records on multiple days. It must also satisfy one of two span criteria: at least 10% of patients span 365 days, or the 75th percentile of patient-level spans is at least 365 days. This fixed rule selects 239 of 6,736 diagnosis codes among 2,094 training patients with diagnosis events. The vocabulary and numeric normalization statistics are also fit only on the training split.

The held-out patients therefore play no role in defining the persistent-code set S . The Raw representation retains the most recent L events, whereas Era+backfill first transforms the complete available history and then takes the same-length suffix:

$$R_i^{\text{Raw}} = \text{suffix}_L(H_i), \quad R_i^{\text{Era}} = \text{suffix}_L(T_{90,S}(H_i)). \quad (2)$$

We use $L = 4096$ for the primary comparison and $L = 16384$ for context-length sensitivity. At $L = 4096$, 53.5% of readmission histories and 41.9% of ICU histories are truncated across all splits.

For each selected diagnosis, duplicate mentions are first summarized by visit day. Consecutive mentions separated by at most 90 days form one condition era. An era with at least two diagnosis points becomes three ordered tokens: a start token, an end token carrying duration in days, and a count-bin token carrying the number of diagnosis points. Single-point eras and all nonselected events remain ordinary events. The full-history ERA+BACKFILL variant transforms the available history before taking the final L positions, so positions freed by compression can admit earlier events.

Unless stated otherwise, the primary ERA+BACKFILL pipeline uses a 90-day gap and $L = 4096$. We refer to its controls consistently as ERA NO-BACKFILL and ERA STRUCTURE-NULL.

We add two controls. ERA NO-BACKFILL first takes the Raw L -event suffix and then transforms it, prohibiting older-event admission. ERA STRUCTURE-NULL exactly preserves the number, timestamps, ordering, base diagnoses, and backfilled positions of ERA+BACKFILL, but replaces special

state tokens by their base diagnosis and removes their state-specific numeric values. It isolates the added explicit state encoding more closely than a Raw comparison, although positions and timing still carry implicit information. For ICU, we additionally evaluate 30- and 180-day era gaps.

The no-backfill comparison is a full-history transformation with backfill versus transformation after Raw truncation. It changes older-event admission and the histories used to construct era boundaries, durations, and counts; only the structure-null control more directly isolates explicit state tokens.

3.3 Models and training

We use a numeric-aware RETAIN-lite model for readmission and a two-layer numeric-aware GRU for ICU transfer. Each event combines a 64-dimensional token embedding, projected temporal features, and projected numeric value/missingness features. Hidden size is 128 and dropout is 0.20. Models use batches of eight, AdamW (10^{-3} learning rate, 10^{-4} weight decay), gradient clipping at 1, and binary cross-entropy with training-set positive weight $n_{\text{negative}}/n_{\text{positive}}$. Training runs for at most 12 epochs; the checkpoint with the best tuning AP is restored, with early stopping after three consecutive epochs without improvement.

Each task–representation pair is trained with seeds 42–46. A logistic calibrator is fitted to tuning logits for each run and applied once to held-out logits [Platt, 1999, Guo et al., 2017]. We average the five calibrated held-out probabilities to obtain

$$\bar{p}_i = \frac{1}{5} \sum_{s=1}^5 \sigma(a_s z_{is} + b_s), \quad (3)$$

where z_{is} is the held-out logit for seed s .

3.4 Evaluation protocol

The primary analysis compares Raw and Era+backfill at $L = 4096$ across five held-out metrics under episode weighting. Secondary analyses examine patient weighting, retraining stability, top-decile agreement, calibration, context and era-horizon sensitivity, and controlled diagnosis repetition.

Development used held-out feedback across 81 variants per task, so the analysis is retrospective. The conditional bootstrap intervals do not account for this adaptive selection. The shared protocol directly measures aggregate performance and agreement in episode prioritization for the fitted pipelines; it does not estimate performance on an untouched confirmatory cohort.

Metrics and estimands. We compute AUROC, average precision (AP), Brier score, log loss, and precision among the highest 10% predicted risks. AP is the recall-increment-weighted mean of precision along the precision–recall curve. AUROC measures ranking across thresholds; AP emphasizes positive-case retrieval; Brier score and log loss assess probability quality; and top-10% precision measures retrieval at a fixed top-decile capacity. We orient every paired effect so that positive favors Era+backfill:

$$\Delta_m = \begin{cases} m(\text{Era}) - m(\text{Raw}), & \text{higher is better,} \\ m(\text{Raw}) - m(\text{Era}), & \text{lower is better.} \end{cases} \quad (4)$$

The primary estimand treats episodes equally but resamples patients as clusters: each bootstrap sample draws patients with replacement and retains all their episodes. We use 10,000 paired replicates and percentile 95% CIs. A sensitivity estimand assigns each patient the same total weight, $w_i = 1/n_{g_i}$, where n_{g_i} is that patient’s episode count. Episodes are ranked by predicted risk, and the boundary episode can enter fractionally so the selected total weight is exactly 10%. This estimand reduces the influence of patients with repeated admissions while preserving the episode-level prediction target. These intervals quantify held-out sampling uncertainty conditional on the fitted five-seed ensembles; they do not include model-selection or full-retraining uncertainty.

Retraining and selected-set stability. Under episode weighting, each pipeline selects the $k = \lceil 0.10N \rceil$ highest-risk episodes from a cohort of size N , with deterministic episode-identifier tie-breaking. The capacity is shared, not the probability cutoff. For two selected sets A and B , each of

size k , we define

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad r(A, B) = \frac{|A \setminus B|}{k} = \frac{|B \setminus A|}{k}.$$

Jaccard measures intersection relative to union; the replacement fraction r measures the share of either fixed-capacity set absent from the other. For baseline and perturbed sets, we call this fraction churn and report entered and exited episodes separately, without counting their sum as the replacement rate.

We report metric means and standard deviations across seeds, the number of seeds favoring each representation, and pairwise Jaccard overlap of seed-specific top-10% prediction-episode sets. We also compare the ensemble top-10% sets across Raw and Era+backfill. This distinguishes stability of global performance from stability of selected episodes [Lopez-Martinez et al., 2022]. For matched-size references, we average calibrated probabilities over every one- or two-seed subset. Within a representation, we compare disjoint subsets; across representations, we compare the same seed-index subset. Sizes one and two yield 10 and 15 within-representation pairs and 5 and 10 cross-representation pairs, respectively. These comparisons reuse fitted models and are not independent replicates. The five-seed cross-representation comparison is reported separately. All seeds use the same training patients and split.

A post-hoc descriptive analysis characterizes representation-specific ensemble selections using the repeat count for one candidate diagnosis, the time from its first mention to prediction, and retained Raw sequence length. The candidate is the deterministic diagnosis chosen for the stress-test plan, not a summary of all diagnoses. Its age since first mention is neither a first-to-last documentation span nor a condition-era duration. We report coverage because these characteristics are available only for episodes included in that plan; this profile is distinct from the strictly pre-prediction stress cohort.

Calibration and multiplicity. For the two primary 4,096-event representations in each task, we report 10-bin equal-frequency expected calibration error (ECE), calibration intercept and slope, and reliability curves. We obtain uncertainty with 2,000 patient-cluster bootstrap samples. Intercept 0 and slope 1 are the targets, complementing Brier score and log loss [Van Calster et al., 2019].

The full analysis tracks 12 representation contrasts across five metrics. Holm and Benjamini-Hochberg sensitivity adjustments are applied across the entire family of 60 contrast-metric quantities, not separately within the five ICU state-versus-null comparisons [Holm, 1979, Benjamini and Hochberg, 1995]. These retrospective summaries accompany nominal patient-cluster intervals; complete details appear in the appendix.

Synthetic repetition stress test. For each eligible held-out episode, we deterministically select one diagnosis from the 239-code list that is already present in at least two reconstructed visits. Later visits where that diagnosis is absent are ranked by a stable hash. Nested subsets requested at 0%, 25%, 50%, or 100% of eligible visits receive a duplicate diagnosis event. The requested fraction f refers to eligible visits, not all visits or events in the history. The number of changed visits is $\lceil f n_{\text{eligible}} \rceil$, so realized fractions at 25% and 50% can be materially higher when few visits are eligible. The same selected diagnosis and visit subset are used for Raw and Era+backfill pipelines. Frozen checkpoints, vocabularies, and calibrators produce new predictions; no model is retrained.

We restrict the analysis to episodes whose eligible visits are strictly before prediction time. This leaves 1,286 of 2,189 readmission episodes (58.7%; 685 patients, 157 positives) and 1,028 of 2,037 ICU episodes (50.5%; 570 patients, 37 positives). The stress-test analysis compares the mean absolute change in calibrated probability from the stress run’s own 0% baseline with a 10,000-sample patient-cluster bootstrap. At the 100% level, we also count episodes entering and leaving each pipeline’s top-10% set, recomputed within this eligible cohort rather than the complete held-out cohort.

For each task-pipeline pair, a post-hoc profile selects the $n_{\text{tail}} = \lceil 0.05N \rceil$ largest absolute calibrated probability shifts from the stress run’s own 0% baseline to 100% repetition. We sort by descending $|\Delta p|$, breaking ties by ascending episode identifier, and compare candidate-diagnosis and eligible-visit medians with the remainder. This tail selection is distinct from top-decile prioritization and is conditional on the strict stress cohort.

4 Results

4.1 Primary predictive performance and paired uncertainty

Table 1 reports the primary five-seed ensembles. At $L = 4096$, Era+backfill gives the better observed value in all ten task–metric comparisons. The readmission differences are 0.0043 AUROC, 0.0108 AP, and 1.37 percentage points in P@10; the corresponding ICU differences are 0.0065, 0.0047, and 0.49 points. This point-estimate ranking is consistent across metrics, while every paired patient-cluster interval spans zero. The aggregate table therefore supplies a clear observed ordering but not a decisive representation choice on its own.

Table 1: Primary retrospective held-out results at 4,096 events. Δ is the Era+backfill benefit, oriented so positive is better; intervals use paired patient-cluster bootstrap conditional on the fitted ensembles. Development reused this held-out cohort across 81 variants per task; intervals do not account for adaptive model selection.

Metric	Readmission			ICU transfer		
	Raw	Era	Δ [95% CI]	Raw	Era	Δ [95% CI]
AUROC	.7942	.7985	.0043 [−.0045, .0135]	.7948	.8013	.0065 [−.0167, .0305]
AP	.4097	.4206	.0108 [−.0120, .0268]	.2194	.2241	.0047 [−.0265, .0422]
Brier	.08690	.08602	.00088 [−.00036, .00217]	.03637	.03618	.00018 [−.00058, .00094]
Log loss	.29956	.29693	.00263 [−.00077, .00614]	.14926	.14745	.00181 [−.00119, .00490]
P@10%	.4521	.4658	.0137 [−.0131, .0631]	.1765	.1814	.0049 [−.0204, .0431]

4.2 Component analysis of persistence-state encoding

Table 2 decomposes the two mechanisms. The structure-matched control holds sequence length, timestamps, base diagnoses, and backfilled positions fixed while removing explicit state tokens and values. In ICU transfer, explicit state encoding improves all five observed metrics over this control, and all five nominal paired intervals exclude zero; the differences include 0.0309 AUROC and 0.0416 AP. The full-history transformation with backfill also improves ICU AUROC and log loss relative to transformation after Raw truncation, but that contrast jointly changes era construction and older-event admission. The earliest retained timestamp moves earlier in only 4.2% of ICU and 5.8% of readmission episodes, making the structure-null comparison the more specific evidence for explicit state.

Table 2: ICU component contrasts. Benefits are oriented so positive is better; intervals use paired patient-cluster bootstrap.

Metric	State vs. structure-null	Full history vs. post-truncation
AUROC	.0309 [.0092, .0524]	.0281 [.0102, .0475]
AP	.0416 [.0049, .0908]	.0234 [−.0091, .0571]
Brier	.00112 [.00010, .00219]	.00063 [−.00011, .00137]
Log loss	.00783 [.00393, .01221]	.00430 [.00153, .00717]
P@10%	.0245 [.0046, .0550]	.0147 [−.0100, .0464]

Across 60 retrospective contrast–metric quantities, 12 nominal intervals exclude zero. An approximate bootstrap sign-tail sensitivity yields three quantities below 0.05 after Holm and seven after Benjamini–Hochberg, with both adjustments applied across all 60 quantities. Among the five ICU state-versus-null comparisons, log loss is the only quantity below 0.05 after this family-wide Holm adjustment. These retrospective quantities are sensitivity summaries rather than confirmatory significance tests.

4.3 Context-length and era-horizon sensitivity

At $L = 16,384$, the preferred representation depends on task and metric (Table 3). For readmission, Raw gains 0.0092 AUROC, whereas Era+backfill gains 2.74 percentage points in P@10. For ICU,

Era+backfill gains 0.0335 AP and 0.00075 Brier, while Raw gains 0.0031 AUROC and 0.49 percentage points in P@10. Longer context thus redistributes performance across evaluation objectives rather than uniformly favoring one representation.

The era-horizon analysis shows the same trade-off. Relative to 90 days, the 180-day ICU horizon improves AP by 0.0121 and P@10 by 1.47 points, but reduces AUROC by 0.0256; the 30-day horizon reduces both AUROC and AP by 0.0463 and 0.0640. The selected configuration therefore depends on whether the objective emphasizes global ranking, rare-outcome retrieval, probability quality, or top-decile precision.

Table 3: Context-length and era-horizon sensitivity for five-seed calibrated ensembles. Gap is the condition-era horizon in days; lower Brier and higher values of the other reported metrics are better.

Task	Representation	L	Gap	AUROC	AP	Brier	P@10%
Readm.	Raw	16,384	–	.8034	.4202	.08804	.4384
	Era+backfill	16,384	90	.7942	.4161	.08767	.4658
ICU	Raw	16,384	–	.7805	.2017	.03690	.1814
	Era+backfill	16,384	90	.7774	.2352	.03615	.1765
	Era+backfill	4,096	30	.7550	.1601	.03765	.1569
	Era+backfill	4,096	90	.8013	.2241	.03618	.1814
	Era+backfill	4,096	180	.7757	.2362	.03634	.1961

4.4 Top-decile selection and retraining stability

The representation choice changes which episodes fill the top-decile capacity. At the same fixed capacity, Era+backfill captures 102 versus 99 readmission positives and 37 versus 36 ICU positives. Those modest count gains mask larger changes in composition: the ensemble sets have Jaccard overlap 0.731 for readmission and 0.600 for ICU, with 34 of 219 readmission episodes (15.5%) and 51 of 204 ICU episodes (25.0%) in each set specific to one representation. These count differences describe observed outcome capture, not the benefit of an intervention or the clinical value of the exchanged episodes.

Table 4 supplies like-sized retraining references. Cross-representation means lie between the two within-representation means in both readmission comparisons and single-seed ICU; two-seed ICU cross overlap is slightly higher than either within mean. These descriptive means do not attribute disagreement to representation; the five-seed overlaps above remain separate comparisons. Training-sample and alternative-split stability remain untested.

Table 4: Mean top-decile Jaccard at matched ensemble sizes m . Within pairs use disjoint seed subsets; cross pairs use matched seed indices. Pair counts and dependence are defined in Methods.

Task	m	Raw within	Era within	Cross representation
Readmission	1	.475	.546	.530
	2	.575	.640	.625
ICU transfer	1	.289	.370	.342
	2	.414	.450	.457

Alternative summaries expose where the apparent preference is least stable. Equal-patient P@10 points toward Raw in both tasks, while ICU seed majorities favor Raw for AUROC, AP, and log loss. These local reversals identify the analysis unit and retraining seed as factors that can change the selected pipeline, even though most aggregate summaries favor Era+backfill. Figure 3(a) collects these protocol-dependent point preferences; the paired episode- and equal-patient intervals all include zero.

4.5 Post-hoc characterization of discordant selections

Table 5 characterizes episodes selected by only one primary ensemble, including the outcome counts underlying the net gains in the full selected sets.

The characterized subset covers 22 of 34 Raw-only readmission episodes (64.7%) and 28 of 34 Era-only episodes (82.4%); corresponding ICU coverage is 21 of 51 (41.2%) and 31 of 51 (60.8%). Outcome and patient counts refer to the complete discordant sets, while diagnosis and retained-length summaries refer only to these characterized subsets. Eligibility is not random, and the different coverage rates limit comparisons between groups.

Table 5: Representation-specific top-10% episodes. Episodes are reported as total (patients; positives); n_c is the characterized subset. Mentions and candidate age are median [IQR]. “Raw at cap” is the characterized fraction with 4,096 retained Raw events.

Task	Set	Episodes (patients; pos.)	n_c	Mentions	Candidate age, days	Raw at cap
Readm.	Raw-only	34 (28; 10)	22	6 [3, 21.5]	442 [120, 1,399]	90.9%
	Era-only	34 (23; 13)	28	5.5 [3, 14.2]	143 [65, 461]	82.1%
ICU	Raw-only	51 (49; 5)	21	18 [4, 33]	388 [112, 1,007]	76.2%
	Era-only	51 (46; 6)	31	7 [4, 16.5]	411 [155, 887]	58.1%

The profile shows no consistent candidate-diagnosis signature across tasks. Among readmission episodes, time since first candidate mention is shorter for Era-only than Raw-only selections (median 143 versus 442 days); in ICU, candidate ages are similar (411 versus 388 days), while Era-only selections have fewer candidate mentions (median 7 versus 18). All four groups have median retained Raw sequence length 4,096, but reaching the cap does not reveal uncapped history length. These eligibility-conditioned groups describe discordance without establishing its causes or clinical significance.

4.6 Calibration and controlled-repetition sensitivity

Era+backfill has lower ECE point estimates by 0.0007 for readmission and 0.0010 for ICU, with calibration-slope point estimates 0.073 and 0.039 closer to their target. Closer point estimates do not establish a paired improvement in calibration. Both readmission slope intervals remain above 1, indicating residual miscalibration after tuning-set calibration. Table 6 reports the full diagnostics and intervals; the reliability curves appear in Appendix A.2.

Table 6: Calibration of the primary ensembles. Intervals use 2,000 patient-cluster bootstrap samples; targets are intercept 0 and slope 1.

Task	Representation	ECE	Intercept [95% CI]	Slope [95% CI]
Readmission	RAW	.0226	.354 [.063, .602]	1.270 [1.119, 1.417]
Readmission	ERA+BACKFILL	.0219	.217 [−.049, .446]	1.197 [1.061, 1.335]
ICU transfer	RAW	.0128	.563 [−.129, 1.300]	1.217 [.992, 1.454]
ICU transfer	ERA+BACKFILL	.0118	.439 [−.238, 1.173]	1.178 [.967, 1.431]

Under full requested repetition, Era+backfill reduces the mean absolute probability change by 74% relative to Raw for readmission, whereas its change is 91% larger for ICU (Figure 3(b)). Rank correlations remain above 0.998 and top-decile churn at or below 2.9%, indicating limited global reranking. Appendix Table 12 reports absolute changes at each requested level.

Episode-level membership changes. Table 7 reports one entering and one exiting readmission episode per pipeline and three of each for ICU. Strict eligible capacities are 129 and 103, versus 219 and 204 in the full cohorts. Raw loses one readmission positive selection, while Era+backfill gains one; all ICU exchanges have negative outcomes. These counts cannot support reliable subgroup inference or clinical benefit claims.

Tail changes and interpretation. At the 100% intervention level, the largest 95th-percentile absolute probability change across pipelines was 0.005739. The largest individual change, 0.097974, occurred for ICU Era+backfill and corresponds to approximately 9.8 percentage points. Table 8 characterizes the largest 5% of shifts: eligible-visit medians are higher than in the remainder for every pipeline, while candidate-mention differences vary by task and pipeline.

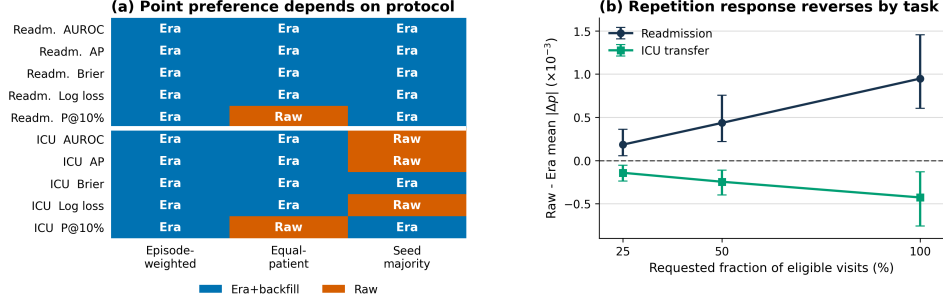


Figure 3: Protocol-dependent selection and repetition sensitivity. (a) Point preferences under episode weighting, equal-patient weighting, and seed majority; colors do not imply significance. (b) Raw minus Era+backfill mean absolute probability change under strict pre-prediction repetition, with paired patient-cluster 95% intervals. Positive means Era+backfill changes less.

Table 7: Top-decile membership changes at 100% requested repetition within strict eligible cohorts. Positive outcomes appear in parentheses; churn is exited episodes divided by k .

Task	Pipeline	Top k	Entered (pos.)	Exited (pos.)	Changed patients
Readm.	Raw	129	1 (0)	1 (1)	2
	Era+backfill	129	1 (1)	1 (0)	2
ICU	Raw	103	3 (0)	3 (0)	6
	Era+backfill	103	3 (0)	3 (0)	4

Table 8: Largest-5%-shift profiles at full repetition in strict stress cohorts. Mentions and eligible visits are medians: tail (remainder). Maximum $|\Delta p|$ is in percentage points (pp); selection is defined in Methods.

Task	Pipeline	n_{tail}	Mentions tail (remainder)	Eligible visits tail (remainder)	Max $ \Delta p $ pp
Readmission	Raw	65	17 (6)	71 (22)	5.577
	Era+backfill	65	5 (7)	48 (22)	3.306
ICU transfer	Raw	52	8.5 (6)	38 (24)	4.065
	Era+backfill	52	15 (6)	59 (23)	9.797

At the 100% requested level, eligible-visit count also determines the number of inserted diagnoses, so visit-count enrichment combines history characteristics with intervention dose.

Baseline Raw histories reach the event cap in 67.7% and 60.0% of the readmission Raw/Era tails, and 61.5% and 65.4% of the ICU Raw/Era tails. For ICU Era+backfill, median time since first candidate mention is 764 days in the tail versus 383 in the remainder. These post-hoc, stress-conditional associations characterize episodes with large shifts, not their causes or clinical significance; tail membership does not imply top-decile entry or exit. Small mean shifts and set churn therefore coexist with larger episode-level sensitivity.

5 Discussion

Cohort performance versus thresholded selection. Global metrics average over a cohort, whereas top-decile selection depends on local ordering. Despite favorable point estimates, 15.5% and 25.0% of selected episodes remain representation-specific after ensembling. Table 4 supplies descriptive retraining references, not attribution. Equal-patient P@10 shows analysis-unit dependence; similar cohort ECE neither explains discordance nor rules out local calibration differences.

Mechanisms of persistence-aware representation. Only 4.2% of ICU and 5.8% of readmission episodes gain an earlier retained timestamp. Structure-null controls associate the clearest ICU gains with explicit state, but the full-history contrast also changes era construction. Context length and

era horizon trade metrics rather than improving them uniformly. Neither aggregate ablations nor the discordant profile in Table 5 explain the clinical causes of selection changes. Persistence state, historical reach, context budget, and era horizon remain distinct design choices.

Task-specific perturbation sensitivity. Controlled repetition is dampened by Era+backfill for readmission and amplified for ICU, precluding one representation-wide robustness claim. The tail profile in Table 8 links larger shifts to specific recorded history characteristics, descriptively rather than causally. Stress membership changes in Table 7 use smaller eligible cohorts than the complete held-out representation comparison, so their rates are not a matched instability baseline. This frozen-pipeline test probes a specified pre-prediction perturbation, not the prevalence or effect of real-world copy-forward or the clinical significance of the affected histories.

Implications for thresholded model selection. For thresholded use, a compact profile should pair use-aligned metrics with patient-cluster uncertainty, calibration, retraining variability, selected-set agreement, and relevant behavioral tests. It should not collapse these into one score: outcome, capacity, and primary metric define the decision, while the other dimensions expose distinct failure modes. The same principle applies whenever continuous scores become finite prioritization sets.

6 Conclusion

Similar cohort-level scores do not imply interchangeable high-risk prioritization. Era+backfill leads every primary point estimate, component controls locate an ICU signal in explicit state encoding, and the five-seed ensembles still disagree at the same prioritization capacity. Paired uncertainty and opposite repetition responses further show why aggregate ranking alone cannot resolve the choice.

The scope is one EHR source, two outcomes, 85 positive ICU held-out episodes, one architecture per task, and a 10% prioritization capacity. Development used held-out feedback across 81 variants per task; bootstrap intervals condition on fitted ensembles, and tuning supports both stopping and calibration. Exact-time events, opaque CARE_SITE codes, and one repetition mechanism leave residual proxy risk. Seeds vary initialization, not sampled training patients, and the post-hoc profile covers only stress-eligible episodes; neither alternative-split stability nor clinical causes of discordance are established.

A locked external cohort and preregistered protocol can test effect magnitudes, selected sets, subgroup behavior, alternative thresholds and splits, and clinical net benefit. The immediate contribution is a reproducible profile that exposes when a small benchmark difference changes who receives priority.

Acknowledgments

The study was supported by the Ministry of Economic Development of the Russian Federation (agreement No. 139-10-2025-034 dd. 19.06.2025, IGK 000000C313925P4D0002).

References

- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010. URL <https://www.jmlr.org/papers/v11/cawley10a.html>.
- Edward Choi, Mohammad Taha Bahadori, Joshua A. Kulas, Andy Schuetz, Walter F. Stewart, and Jimeng Sun. RETAIN: An interpretable predictive model for healthcare using reverse time attention mechanism. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL <https://proceedings.neurips.cc/paper/2016/file/231141b34c82aa95e48810a9d1b33a79-Paper.pdf>.
- Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B. Reitsma, Maarten van Smeden, et al. TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.
- Alexander D’Amour, Katherine Heller, Dan Moldovan, Ben Adlam, Babak Alipanahi, Alex Beutel, Christina Chen, Jonathan Deaton, Jacob Eisenstein, Matthew D. Hoffman, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022. URL <https://jmlr.org/papers/v23/20-1335.html>.
- Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349(6248):636–638, 2015. doi: 10.1126/science.aaa9375.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. doi: 10.2307/4615733.
- Daniel Lopez-Martinez, Alex Yakubovich, Martin Seneviratne, Adam D. Lelkes, Akshit Tyagi, Jonas Kemp, Ethan Steinberg, N. Lance Downing, Ron C. Li, Keith E. Morse, Nigam H. Shah, and Ming-Jun Chen. Instability in clinical risk stratification models using deep learning. In *Proceedings of the 2nd Machine Learning for Health Symposium*, volume 193 of *Proceedings of Machine Learning Research*, pages 552–565. PMLR, 2022. URL <https://proceedings.mlr.press/v193/lopez-martinez22a.html>.
- Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, Ben Glocker, Fabian Isensee, Jens Kleesiek, Michal Kozubek, et al. Metrics reloaded: Recommendations for image analysis validation. *Nature Methods*, 21(2):195–212, 2024. doi: 10.1038/s41592-023-02151-z.
- MEDS Working Group, Bert Arnrich, Edward Choi, Jason A. Fries, Matthew B. A. McDermott, Jungwoo Oh, Tom J. Pollard, Nigam Shah, Ethan Steinberg, Michael Wornow, and Robin van de Water. Medical event data standard (MEDS): Facilitating machine learning for health. In *ICLR 2024 Workshop on Learning from Time Series for Health*, 2024. URL <https://openreview.net/pdf?id=IsHy2ebjIG>.
- John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Alexander J. Smola, Peter L. Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.
- Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432, 2015. doi: 10.1371/journal.pone.0118432.

- Stanford Shah Lab. EHRSHOT benchmark repository: Task definitions and labeler source. GitHub repository, 2026. URL <https://github.com/som-shahlab/ehrshot-benchmark>. README and task labeler source, accessed 2026-08-21.
- J. Daryl Thornton, Jesse D. Schold, Lokesh Venkateshaiah, and Brian Lander. Prevalence of copied information by attendings and residents in critical care progress notes. *Critical Care Medicine*, 41(2):382–388, 2013. doi: 10.1097/CCM.0b013e3182711a1c.
- Ben Van Calster, David J. McLernon, Maarten van Smeden, Laure Wynants, and Ewout W. Steyerberg. Calibration: The Achilles heel of predictive analytics. *BMC Medicine*, 17(1):230, 2019. doi: 10.1186/s12916-019-1466-7.
- Max Westphal and Werner Brannath. Improving model selection by employing the test data. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 6747–6756. PMLR, 2019. URL <https://proceedings.mlr.press/v97/westphal19a.html>.
- Michael Wornow, Rahul Thapa, Ethan Steinberg, Jason A. Fries, and Nigam H. Shah. EHRSHOT: An EHR benchmark for few-shot evaluation of foundation models. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/d42db1f74df54cb992b3956eb7f15a6f-Paper-Datasets_and_Benchmarks.pdf.
- Michael Wornow, Suhana Bedi, Miguel Angel Fuentes Hernandez, Ethan Steinberg, Jason Alan Fries, Christopher Ré, Sanmi Koyejo, and Nigam H. Shah. Context clues: Evaluating long context models for clinical prediction tasks on EHRs. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/2a246ec4437808525a088abae744c2d3-Paper-Conference.pdf.

A Additional Experimental Details and Results

Split integrity and temporal boundary. Nine saved checks cover expected split values, duplicate examples, patient overlap, one-split-per-patient, one-split-per-row, agreement with official subject assignments, cross-source consistency, row mapping, and label coverage. All passed with zero issues, establishing the integrity of split and row assignments. The final boundary is $t_{\text{event}} \leq t_{\text{prediction}}$; these checks do not rule out all temporal proxies or semantic leakage through opaque codes. Table 9 gives the audited cohort counts.

Table 9: Episode, patient, and positive-outcome counts by official split.

Task	Split	Episodes	Patients	Positives
Readmission	Train	2,608	1,337	370
	Tuning	2,206	1,191	281
	Held-out	2,189	1,190	260
ICU transfer	Train	2,402	1,306	113
	Tuning	2,052	1,157	92
	Held-out	2,037	1,154	85

Persistence whitelist. The train split contains 2,295 subjects, of whom 2,094 have diagnosis events, 459,789 diagnosis events, and 6,736 unique diagnosis codes. The fixed rule in Section 3 selects 239 codes.

A.1 All ensemble metrics

Table 10 reports every final task–representation ensemble, including the component and sensitivity variants summarized in the main text.

Table 10: Five-seed calibrated ensemble results for every final task–representation pair.

Task	Representation	AUROC	AP	Brier	Log loss	P@10%
Readm.	Raw 4096	.7942	.4097	.08690	.29956	.4521
	Era+backfill 4096	.7985	.4206	.08602	.29693	.4658
	Era no-backfill 4096	.8007	.4052	.08710	.29915	.4703
	Era structure-null 4096	.7928	.4126	.08688	.29996	.4566
	Raw 16384	.8034	.4202	.08804	.30562	.4384
	Era+backfill 16384	.7942	.4161	.08767	.30342	.4658
ICU	Raw 4096	.7948	.2194	.03637	.14926	.1765
	Era+backfill 4096	.8013	.2241	.03618	.14745	.1814
	Era no-backfill 4096	.7732	.2007	.03682	.15176	.1667
	Era structure-null 4096	.7704	.1825	.03731	.15528	.1569
	Raw 16384	.7805	.2017	.03690	.15305	.1814
	Era+backfill 16384	.7774	.2352	.03615	.15053	.1765
	Era+backfill gap30	.7550	.1601	.03765	.15565	.1569
	Era+backfill gap180	.7757	.2362	.03634	.14995	.1961

A.2 Calibration curves

Figure 4 visualizes the calibration summaries reported in Table 6 without adding a separate selection criterion.

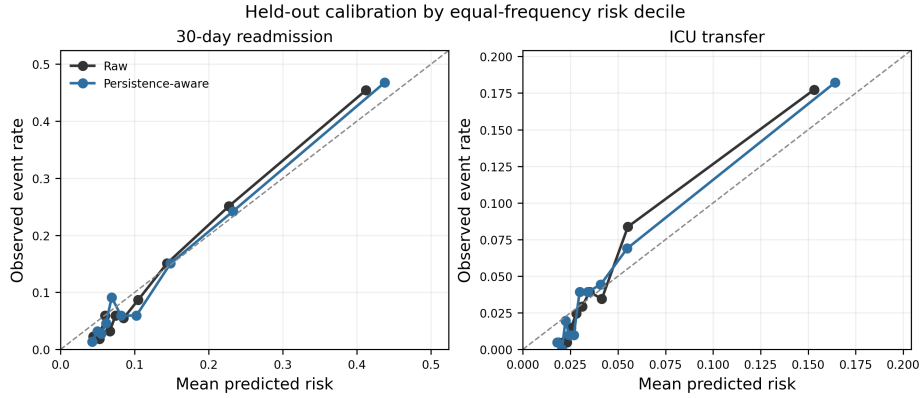


Figure 4: Equal-frequency reliability curves for the four primary ensembles. Points plot observed outcome rate against mean predicted risk in ten held-out bins; the diagonal denotes perfect calibration. Bin-level intervals are not shown. Table 6 reports ECE, intercept, slope, and patient-cluster intervals.

A.3 Seed-level summary

Table 11 separates ensemble-level comparisons from variation across the five complete retraining runs.

Table 11: Mean $\pm$ SD across five independently trained seeds for the primary pipelines. Direction counts are Era+backfill/Raw/tie; lower is better for Brier and log loss.

Task	Metric	Raw	Era+backfill	Direction count
Readm.	AUROC	.7838 $\pm$.0105	.7866 $\pm$.0102	3/2/0
	AP	.3834 $\pm$.0273	.3931 $\pm$.0142	3/2/0
	Brier	.08922 $\pm$.00201	.08805 $\pm$.00149	4/1/0
	Log loss	.30629 $\pm$.00538	.30282 $\pm$.00526	4/1/0
	P@10%	.4183 $\pm$.0306	.4384 $\pm$.0168	3/1/1
ICU	AUROC	.7484 $\pm$.0442	.7636 $\pm$.0164	2/3/0
	AP	.1833 $\pm$.0396	.1963 $\pm$.0252	2/3/0
	Brier	.03762 $\pm$.00081	.03733 $\pm$.00068	3/2/0
	Log loss	.15527 $\pm$.00486	.15285 $\pm$.00335	2/3/0
	P@10%	.1480 $\pm$.0181	.1706 $\pm$.0220	3/1/1

A.4 Multiplicity sensitivity

Twelve of 60 nominal 95% CIs exclude zero. Three Holm-adjusted and seven Benjamini–Hochberg-adjusted bootstrap sign-tail quantities are below 0.05. Both adjustments use the full family of 60 contrast–metric quantities. Each quantity is twice the smaller bootstrap sign fraction with a Monte Carlo +1 correction, based on 10,000 replicates. Its exact null calibration is not guaranteed, the family was retrospective, and these adjusted values do not provide confirmatory FWER or FDR control.

A.5 Synthetic repetition results

Table 12 gives the requested repetition levels and paired changes underlying Figure 3(b).

Table 12: Mean absolute probability change under strict pre-prediction synthetic repetition. Fractions are requested values; the number of changed visits is ceiling-rounded. The difference is Raw minus Era+backfill; positive values mean Era+backfill changes less.

Task	Requested fraction	Raw	Era+backfill	Difference [95% CI]
Readm.	25%	.000408	.000224	.000184 [.000057, .000361]
Readm.	50%	.000710	.000274	.000436 [.000219, .000755]
Readm.	100%	.001275	.000327	.000948 [.000604, .001454]
ICU	25%	.000099	.000241	−.000142 [−.000236, −.000055]
ICU	50%	.000219	.000465	−.000245 [−.000397, −.000110]
ICU	100%	.000471	.000898	−.000427 [−.000756, −.000131]

The initial stress inference reproduces its own 0% baseline exactly by construction. Compared with the separately saved main-run baseline, maximum absolute differences are 0.0050 in logit and 0.000344 in calibrated probability, consistent with rebuilding inference on a different device. All reported deltas use the within-stress 0% baseline.

Artifact availability. Code, configurations, publication-safe aggregate tables, and execution instructions are available at <https://github.com/LvovDmitro/ehrshot-state-or-space>. `python paper/reproduce_public.py` verifies publication-safe aggregate tables and regenerates the paper figures offline, without clinical credentials. It does not retrain models or recompute row-level bootstrap or profile analyses; those require credentialed EHRSHOT access and the restricted inputs. EHRSHOT source records, subject identifiers, row-level predictions, and model checkpoints are not redistributed. The exact upstream revision and checksum of the local MEDS archive were not retained; reproduction is anchored to cohort counts, split audits, and derived-artifact checksums.

The maintained release preserves original authorship and states its licensing scope in `LICENSING.md`: no blanket software license has been assigned. EHRSHOT and MEDS upstream code retain their own Apache-2.0 licenses, which do not license this release or the clinical data. Access to EHRSHOT records remains governed by the benchmark’s Credentialed Health Data License.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The claims match the two-task evaluation and distinguish observed rankings from paired uncertainty.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 6 states the data, selection, architecture, threshold, proxy, and validation scope.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Methods and appendices specify training and evaluation; `paper/reproduce_public.py` verifies public aggregate tables and regenerates figures offline. Retraining and row-level recomputation require credentialed EHRSHOT access and restricted inputs; source-provenance limitations are stated. Code and instructions are available at <https://github.com/LvovDmitro/ehrshot-state-or-space>.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in

some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Public code and aggregates at https://github.com/LvovDmitro/eh_rshot-state-or-space support offline aggregate verification and figure regeneration. This combined item remains No because full retraining requires credentialed EHRSHOT records and restricted derivatives; public availability does not constitute a blanket software license.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 3 and A report splits, preprocessing, architectures, optimization, seeds, calibration, metrics, and inference.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Five retraining seeds, paired patient-cluster intervals, and multiplicity sensitivity are reported.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Model dimensions and optimization are reported, but historical hardware and runtime totals are unavailable.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The study follows benchmark data-use terms, redistributes no restricted records, and limits clinical claims.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Discussion identifies transparent prioritization as a benefit and redirected follow-up as a risk, with monitoring safeguards.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: No deployable clinical model or new sensitive dataset is released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Original authorship is preserved, EHRSHOT data terms and upstream EHRSHOT/MEDS Apache-2.0 licenses are distinguished, and LICENSING.md states that no blanket software license has been assigned to the maintained release.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The public repository at <https://github.com/LvovDmitro/ehrs-hot-state-or-space> documents source, analysis, configurations, environment, manifests, aggregate outputs, offline reproduction, and credentialed retraining requirements; `LICENSING.md` states the absence of a blanket license grant.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study involved no crowdsourcing or participant interaction.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The study used a deidentified benchmark and introduced no new participant interaction.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs were not an experimental or methodological component.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.